

VANGOGH IMAGE COLOURISATION

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Abstract

The use of generative adversarial networks, also often referred to as GAN as a solution to image-to-image translation problems. These ‘problems’ for the scope of this paper are converting black and white images to coloured images. Our reasons to use GAN are fairly simple, these networks allow us to train our model with a loss function and allow the training to be completely self-supervised without any interference required by the user. There already are some APIs like chroma and DeOldify that can do this, but they are behind paywalls and their main focus in the training process was to make a good CNN-based generator and supervised learning while the focus of our project is to work hand in hand with a CNN-generator and the GAN network.

1. Introduction

Image-to-image translations or any other computer vision-related problems require you to turn the image into a more

processable format. The default and easiest way are usually to turn the image into a vector using the RGB (or the OpenCV default of BGR) colour space where each pixel has a red, green and blue value inside an array itself which defines the respective pixel’s colour and then train the model to predict the values of these numbers. This is a very standard approach where nothing can go wrong, but what it leaves us with is having to train a model to predict a rank-3 array where every channel can have 256 values (0 to 255), which means that there will be around 256^3 or about sixteen million cases in total to predict from. This paper will later discuss how this number is minimised to make our training more efficient.

Since this particular solution will be using generative adversarial networks, which will be needing a generator and a discriminator. The generator will be based on the U-Net architecture since it’s the best way one can maintain the image’s quality to some extent while colourising it. This solution will also be using a Markovian discriminator which will help with training

with datasets of as many as 10,000 images per epoch.

The loss functions are what made GAN such a good idea for the solution to this problem. The loss function used will be one which was originally designed for general image-to-image translation cases and minimise it a little for our convenience.

Finally, this paper will also try to use some pre-existing libraries like *fastai* and use the Res-Net architecture to improve our results.

In Summary, this paper presents

1. An efficient approach to image colourisation without any sort of user intervention.
2. Models for generators and discriminators for adversarial network training.
3. Understanding the loss function and its importance.
4. Improving our baseline model using pre-trained models.

The final presentable code and format for the model will be available at: <https://colab.research.google.com/drive/13PNc80xHbFpDY-OjVmJEtUjLhTlNtdDm?usp=sharing>

2. Related Work

Even though this is one of the more popular image-to-image transition problems that have a good number of solutions to scroll through, most of the popular ones use the same approach differently. This is the

encoder-decoder model approach which is one of the more basic forms of a Convolutional Neural Network (CNN).

The most popular solution to this problem to date has been Richard Zhang's model [3] (Figure 1) which was published in 2016. This is the model which inspired the ones that came after it. The approach this model used was simple, it used an encoder to identify the objects in an image and the decoder to fill in the suitable colours for every object as it learnt from the dataset it was given. This model was very successful in its job and also inspired this paper to use a similar CNN-based model with some more additions as the generator for our GAN network.

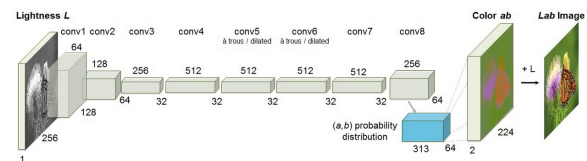


Figure 1: A diagrammatical representation of Richard Zhang's Model

Another model that took heavy inspiration from Richard Zhang's model was Satoshi Iizuka's Model [2] (Figure 2). This model also used a CNN-based encoder-decoder approach but where it shined is that the network had two layers instead of one. One of the layers downsampled the images to identify objects while the other downsampled to identify patterns and margins giving a sharper and clearer output. This model is what inspired the GAN approach, where instead of two layers, two separate models work together (or against each other) to reach the common goal.

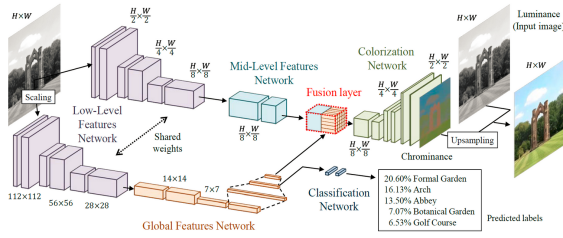


Figure 2: A diagrammatical representation of Satoshi Iizuka's Model

3. Proposed System

In this section is a discussion about the approaches and means used in the solution proposed from a theoretical perspective.

3.1. RBG vs La*b*

Since the scope of the proposed model is only to convert black and white images to coloured images, which means there will be a grayscale image as the input available to the agent. This means using a colour space that can use this input can make the training more efficient. This is where the La*b* colour space comes in. La*b* stands for Lightness, a* channel and b* channel (values -128 to +127 for both of these channels). To make this easier to understand, instead of the Red, Green and Blue layers the RGB Colour space uses, La*b* uses three layers of the grayscale image (lightness), a red-green channel (a* channel) and a blue-yellow channel (b* channel). (Figure 3) shows the difference between the colour spaces.

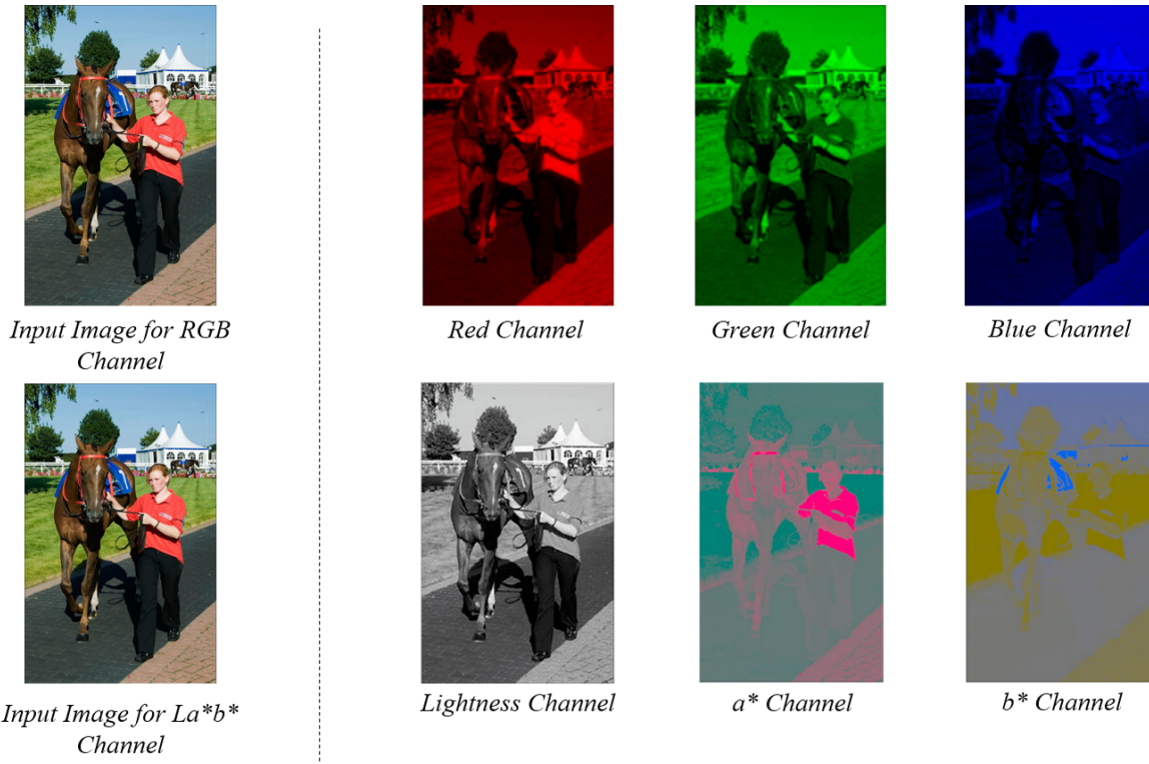


Figure 3: Showing the difference in channels in the RGB and La*b* format (a* and b* channels are coloured just for demonstration purpose). In the RGB Format, we have to predict the three given channels whereas in La*b*, we just need to predict the later two.

What this does is that instead of a rank-3 matrix, now the model trains for a rank-2 matrix, which brings down the number of cases all the way to a little over sixty-five thousand which is a significant jump. This will help in making the training process a lot faster while still not changing the approach within the training itself.

3.2. Model for Generator

The most popular solution to this problem uses an encoder-decoder network approach [5]. How this approach works is the input from the input layer is downsampled repeatedly until reaching a bottleneck point, where upsampling starts till reaching our output. For the problem we discuss, a lot of information needs to be shared between the input and the output. For this, the connection is given means to go through or skip some of the layers and connections which allows information like different objects and edges for colours to circumvent through.

This forms a U-shape and thus is called U-Net. (Figure 4) shows the simplified structure of a U-Net architecture for demonstration. The black arrows show the normal flow of information in the encoder-decoder approach. In contrast, the blue arrows are the ‘skip paths’ that were talked about in the U-Net architecture and they give information to upsampling layers for more accurate results [6].

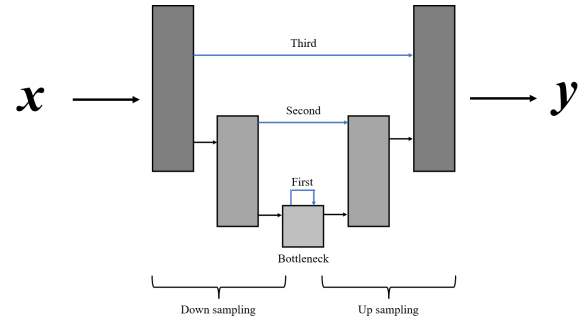


Figure 4: Simplified Structure of U-Net architecture for demonstration and understanding purposes.

3.3. The Loss Function

The objective of the conditional GAN can be expressed by

$$\mathcal{L}_{cGAN}(A, B) = \mathbb{E}_{i,o} [\log B(i,o)] + \mathbb{E}_{i,o} [\log (1 - B(i, A(i,n)))]$$

Considering i as the greyscale image, n as noise for the A and o as the rank-2 array output we expect from the generator (The a^* and b^* colour channels). Also, A is the generator model and B is the discriminator. Note that this is an adversarial network where A and B try to maximise themselves and minimise the other at the same time [4].

This error function is still functional and will give reasonable output but to make the results more accurate, one can add it to the mean absolute error of predicted colours against the actual colours [7].

$$\mathcal{L}_{L1}(A) = \mathbb{E}_{i,o,n} [\|o - A(i,n)\|_1]$$

If mean absolute error loss or L1 loss by itself is used, the model still colourises images with amazing recognition of lines and patterns but will also use colours like

“grey” or “brown” because when it cannot choose between colours, it takes the average and uses these colours that can reduce the L1 loss a lot. Also, the L1 Loss is preferred over or mean squared error because it reduces the production of greyish images. So, the final loss function is

$$A^* = \arg \min_A \max_B \mathcal{L}_{cGAN}(A, B) + \lambda \mathcal{L}_1(A)$$

where λ is a variable that changes for every bot depending on how much contribution lies with what loss in L1 and L2.

such a model for the task of colourisation is great because local features and details are the life of the image in some cases and also help with accuracy and quality. for the solution as a whole in some cases. Here, the model’s output shape is 60 by 60. The actual patch size is obtained when one calculates the output for every one of 3600 outputs which in our case will be 40 by 40.

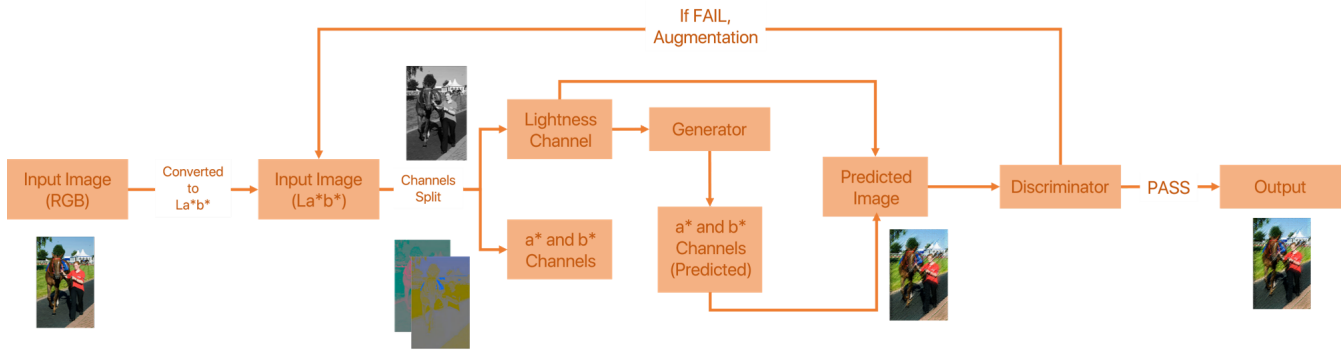


Figure 5: Flow Diagram of System Working

3.4. Model for Discriminator

The model this paper will be using for the discriminator is the *Markovian discriminator* model or *PatchGAN*. [1]

In simple terms, what PatchGAN does is that instead of processing images as a whole, it does it in small patches. In a vanilla discriminator, there is a simple one output which signifies the discriminator’s take on the image being real or fake. In a patch discriminator, the model outputs the same for every patch or part of, for example, 40 by 40 pixels of the input image. Using

4. Implementation

With all the components discussed, the proposed system will have the composition as shown in (Figure 5) where this paper will be using a Generator based on U-Net architecture as previously discussed (section 3.2) and a Discriminator following the Markovian Model (section 3.4). This solution will also be using the COCO 2017 dataset for training for two reasons, it’s big enough for the model training, and it’s easily available to

download via the command line in Google Colaborate where the code is implemented.

The system is really simple, the dataset will be divided into a training set and a validation set-like convention, where the validation set is given to the discriminator to learn what real images look like and the generator gets the training set to work with and generate outputs that get repeatedly better until it can fool the discriminator approximately half the times into choosing our generated output as the real image. The weights at this point of the training will then be saved for future use since the training took about 21 days on Google Colaborate.

Some utility functions will also be implemented for basic counting and tracking

purpose in order to study the results obtained.

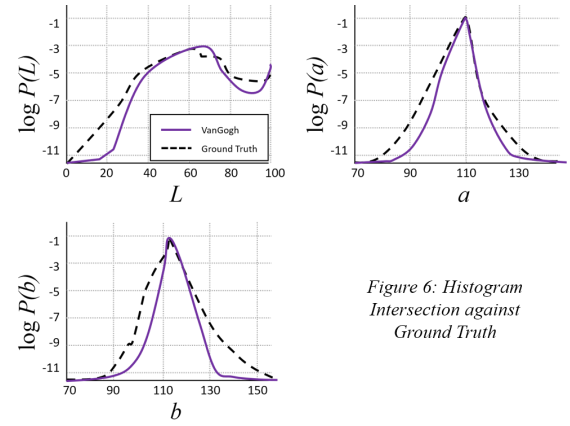


Figure 6: Histogram Intersection against Ground Truth

5. Results

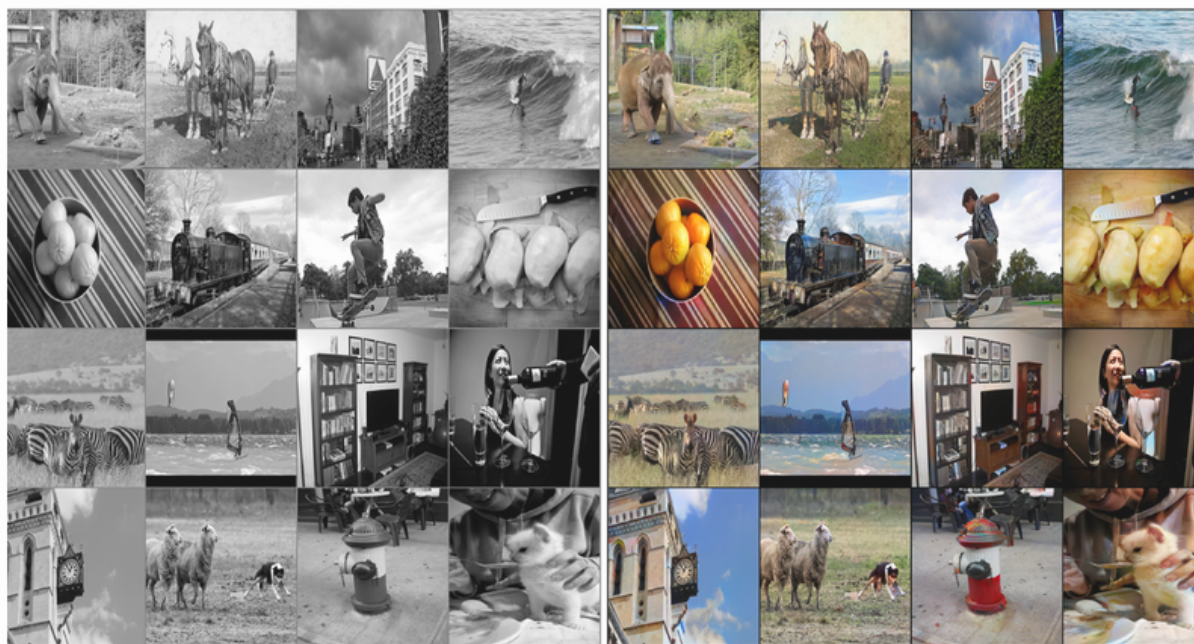


Figure 7: This figure shows the results generated by the solution proposed in this paper. The images on the left represent the input images while the images on the right represent the outputs for the same images.

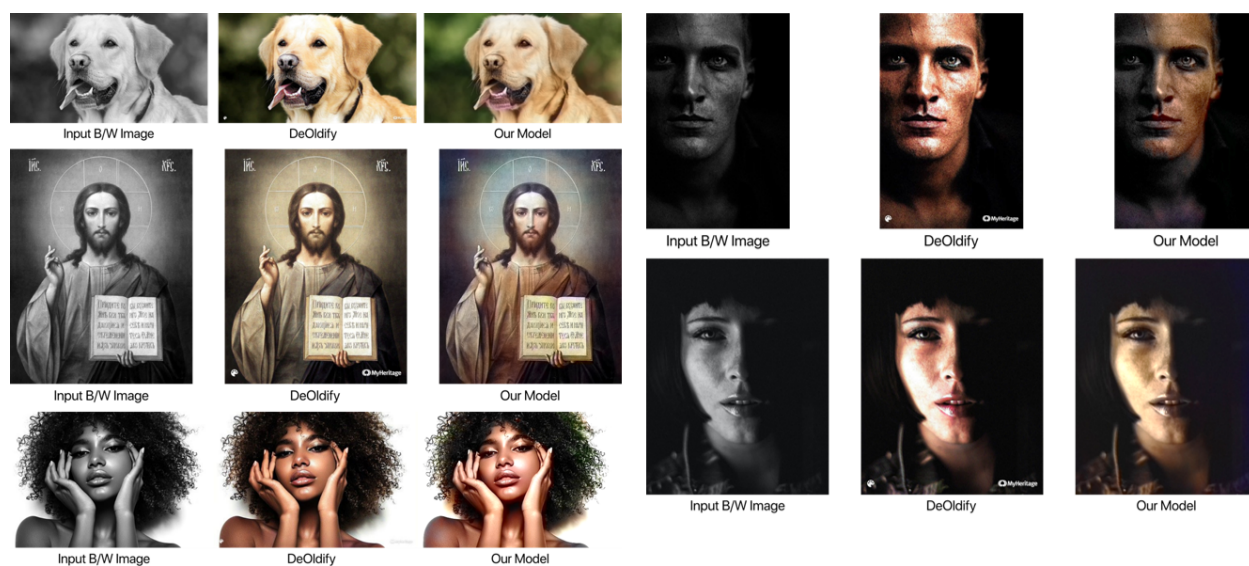


Figure 8: Comparison with an existing Model, DeOldify

As shown by the results, the outputs generated by VanGogh have been successful and comparable with DeOldify, which is one of the leading engines to colourise images in today's cyberspace. In some cases, VanGogh even outclasses DeOldify because of the problem with DeOldify setting higher exposure or oversaturating images. However, in most cases, VanGogh is outclassed by DeOldify due to the years of training that model has gone through. Nonetheless, it shows that VanGogh is a viable solution to the image colourisation problem.

6. Conclusion

It is thus evident that using generative adversarial networks for image colourisation is not only a viable but also a fast and efficient solution to the image colourisation problem. There still were some unsolved issues by the end with images like paintings where the model would fill in similar colours perhaps mistaking too many colours as shadows and thus shades of the same colour, but such imperfections can be easily fixed when trained on a bigger and dedicated dataset. Apart from that, all images retained quality and sharpness after going through colourisation and also the colours were bright and beautiful, making it hard to believe the image was ever colourless, which is a massive success on the record. The results were also obtained relatively fast for an unsupervised model. Overall, it was thus proven that GANs are one of the best ways to implement a colourisation AI.

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7. References

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